Breast Cancer Detection Using Support Vector Machine: A Study on a Custom Synthetic Regional Dataset of Jharkhand

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***Abstract*—Breast cancer is a major health concern, and early identification of potentially malignant tumors is important for timely medical evaluation. This paper presents a machine learning-based classification approach using Support Vector Machine (SVM) to classify breast tumor records into benign and malignant categories. The study uses a custom synthetic dataset containing 569 records representing six regions of Jharkhand: Ranchi, Jamshedpur, Dhanbad, Dumka, Hazaribagh, and Bokaro. The dataset contains five input features: age, tumor size, lymph node status, family history, and Body Mass Index (BMI). An SVM classifier with a Radial Basis Function (RBF) kernel is used as the primary model, while a Decision Tree classifier is used as a baseline for comparison. The data is divided into training and testing sets using an 80:20 split. StandardScaler is applied to the input features before SVM training. The SVM model achieves an accuracy of 88.60%, while the Decision Tree baseline achieves 84.21%. The results indicate that the SVM classifier performs better than the selected baseline on this dataset. Since the dataset is synthetic, the findings represent an experimental machine learning evaluation and do not establish clinical effectiveness.**

***Index Terms*—Support Vector Machine, Breast Cancer Classification, Machine Learning, RBF Kernel, Decision Tree, Synthetic Dataset, Jharkhand.**

I. Introduction

Machine Learning (ML) is increasingly used in healthcare research for identifying patterns in data and supporting classification tasks. Breast cancer is one of the important areas where computational methods can be explored to classify tumor-related records into benign and malignant categories.

Support Vector Machine (SVM) is a supervised learning algorithm used for classification. It identifies a decision boundary that separates data points belonging to different classes while maximizing the margin between them [3]. The kernel trick allows SVM to model non-linear decision boundaries without explicitly transforming the input data into a higher-dimensional feature space.

This study implements an SVM classifier using a Radial Basis Function (RBF) kernel on a custom synthetic dataset representing six regions of Jharkhand. A Decision Tree classifier is included as a baseline to compare classification performance.

The main contributions of this study are:

- Preparation and analysis of a custom synthetic dataset containing 569 records representing six regions of Jharkhand.
- Implementation of an SVM classifier using an RBF kernel.
- Evaluation using accuracy, precision, recall, F1-score, and a confusion matrix.
- Comparison of the proposed SVM model with a Decision Tree baseline.

II. Literature Review

Several studies have explored Support Vector Machine techniques for medical diagnosis and other classification problems.

Zhou et al. (2022) [1] discussed the use of SVM-based methods for mining electronic medical records and identifying patterns relevant to disease diagnosis.

Abdelfattah et al. (2023) [2] presented a privacy-preserving medical diagnosis approach based on Support Vector Machine techniques.

Guido (2024) [3] provided an overview of advancements in Support Vector Machine methods, including kernel-based approaches for classification.

Huynh et al. (2024) [4] investigated the application of SVM in rotating machinery fault detection using feature extraction techniques.

These studies provide background on the application of SVM across classification and pattern recognition tasks. The present work focuses on a smaller experimental classification study using a custom synthetic dataset.

III. Methodology

The proposed system uses supervised machine learning to classify records into two categories: benign and malignant. The workflow consists of dataset loading, feature selection, label encoding, train-test splitting, feature scaling, model training, and performance evaluation.

TABLE I
FEATURES OF THE SYNTHETIC JHARKHAND DATASET

Feature	Description	Type
Patient_ID	Unique record identifier	Text
Region	Region of Jharkhand	Categorical
Age	Age of the record	Numerical
Tumor_Size_mm	Tumor size in mm	Numerical
Lymph_Node_Status	Lymph node status	Binary
Family_History	Family history indicator	Binary
BMI	Body Mass Index	Numerical
Diagnosis	Benign/Malignant	Target

A. Support Vector Machine

Support Vector Machine is a supervised learning algorithm that constructs a decision boundary, also called a hyperplane, to separate different classes.

For a linearly separable dataset, the decision boundary can be represented as:

$$w^T x + b = 0 \tag{1}$$

where w represents the weight vector, x represents the input feature vector, and b is the bias term.

B. Radial Basis Function Kernel

In this study, the RBF kernel is used to handle possible non-linear relationships between input features.

The RBF kernel is defined as:

$$K(x_i, x_j) = \exp\left(-\gamma \|x_i - x_j\|^2\right) \tag{2}$$

where x_i and x_j are input feature vectors and γ controls the influence of individual training samples.

The SVM model uses the following parameters:

- Kernel: RBF
- Regularization parameter (C): 1.0
- Gamma: scale

C. Decision Tree Baseline

A Decision Tree classifier is used as the baseline model. It performs classification by recursively dividing the feature space according to selected feature conditions.

The Decision Tree model is trained and evaluated using the same training and testing records as the SVM model.

IV. DATASET DESCRIPTION

A. Custom Synthetic Dataset

The experiment uses a custom synthetic dataset containing 569 records. The records represent six regions of Jharkhand: Ranchi, Jamshedpur, Dhanbad, Dumka, Hazaribagh, and Bokaro.

The dataset contains demographic and tumor-related features. The target variable, *Diagnosis*, contains two classes: Benign and Malignant.

The dataset contains no missing values or duplicate records in the inspected CSV file.

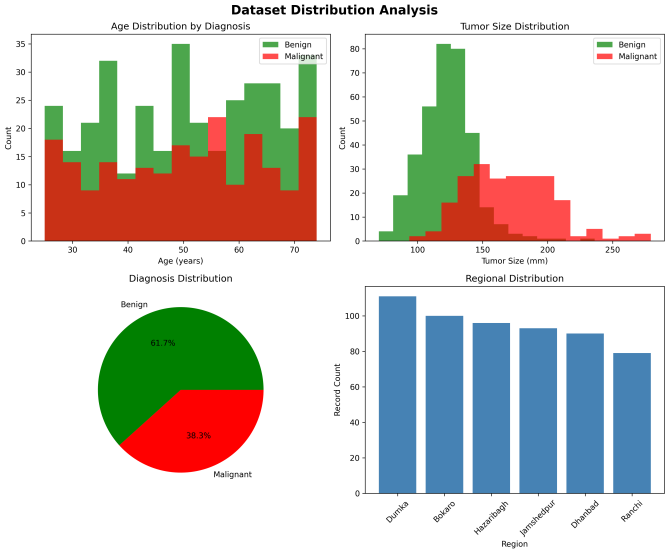


Fig. 1. Distribution of records across selected dataset features and regions.

The dataset contains 351 Benign records and 218 Malignant records. Patient_ID and Region are retained for record identification and dataset analysis. They are not used as input features for model training.

The five features used for classification are:

$$X = \{Age, TumorSize, LymphNode, FamilyHistory, BMI\} \tag{3}$$

The target variable is:

$$y = Diagnosis \tag{4}$$

B. Dataset Distribution

Fig. 1 presents the distribution of the records across the dataset's regions and selected clinical features.

V. IMPLEMENTATION

The proposed system was implemented in Python using the Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn libraries. The dataset was loaded from a CSV file and preprocessed by encoding the target variable into numerical values, where Benign is represented by 0 and Malignant by 1.

Five input features were selected: Age, Tumor Size, Lymph Node Status, Family History, and BMI. The dataset was divided into training and testing subsets using an 80:20 ratio. StandardScaler was applied to normalize the input features for SVM classification.

The SVM classifier was implemented using an RBF kernel with $C = 1.0$ and $\gamma = \text{scale}$. A Decision Tree classifier was also trained as a baseline model using the same training and testing split.

The following listing presents the core implementation of the proposed classification system.

```python
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.svm import SVC
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score

# Load dataset
df = pd.read_csv("Jharkhand_Breast_Cancer_Dataset.
    csv")

# Feature selection and target encoding
X = df[['Age', 'Tumor_Size_mm', 'Lymph_Node_Status',
        'Family_History', 'BMI']]

y = df['Diagnosis'].map({
    'Benign': 0,
    'Malignant': 1
})

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Feature scaling
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# Train SVM classifier
svm_model = SVC(
    kernel='rbf',
    C=1.0,
    gamma='scale'
)

svm_model.fit(X_train_scaled, y_train)
svm_pred = svm_model.predict(X_test_scaled)

# Train Decision Tree baseline
dt_model = DecisionTreeClassifier(random_state=42)
dt_model.fit(X_train, y_train)
dt_pred = dt_model.predict(X_test)

# Model evaluation
svm_accuracy = accuracy_score(y_test, svm_pred)
dt_accuracy = accuracy_score(y_test, dt_pred)

print("SVM Accuracy:", svm_accuracy * 100)
print("Decision Tree Accuracy:", dt_accuracy * 100)
```

Listing 1. Core Implementation of SVM-Based Classification

The performance of both classifiers was evaluated using accuracy, precision, recall, F1-score, and the confusion matrix. The generated visualizations were used for further analysis of dataset distribution, classification performance, feature correlation, and the SVM decision boundary.

VI. Results and Analysis

The SVM and Decision Tree classifiers were evaluated using the same 80:20 training and testing split. The evaluation metrics include accuracy, precision, recall, and F1-score.

A. SVM Model Performance

The SVM classifier achieved an accuracy of 88.60% on the test dataset. Its confusion matrix is presented in Fig. 2.

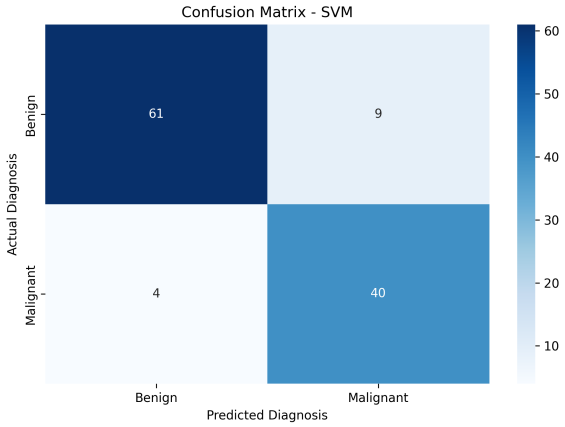


Fig. 2. Confusion matrix of the SVM classifier.

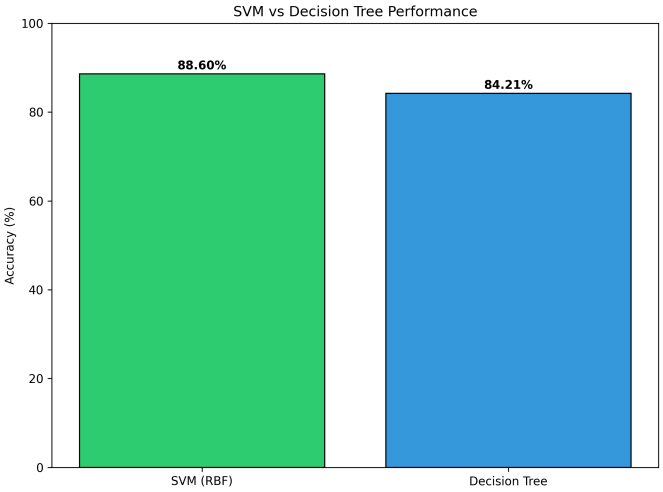


Fig. 3. Accuracy comparison between SVM and Decision Tree.

The confusion matrix contains 61 true negatives, 9 false positives, 4 false negatives, and 40 true positives. Therefore, the model correctly classified 101 out of 114 test records.

B. Model Comparison

The SVM model is compared with the Decision Tree baseline. The comparison is based on the same test dataset.

C. Feature Correlation and Decision Boundary

The feature correlation heatmap in Fig. 4 presents the relationships between the selected numerical and encoded features.

The decision boundary shown in Fig. 5 provides a two-dimensional visualization of the SVM classifier. It uses Age and Tumor_Size_mm for visualization and does not represent the complete five-feature classification model.

VII. Comparative Analysis

The performance of the SVM classifier is compared with the Decision Tree baseline using four evaluation metrics.

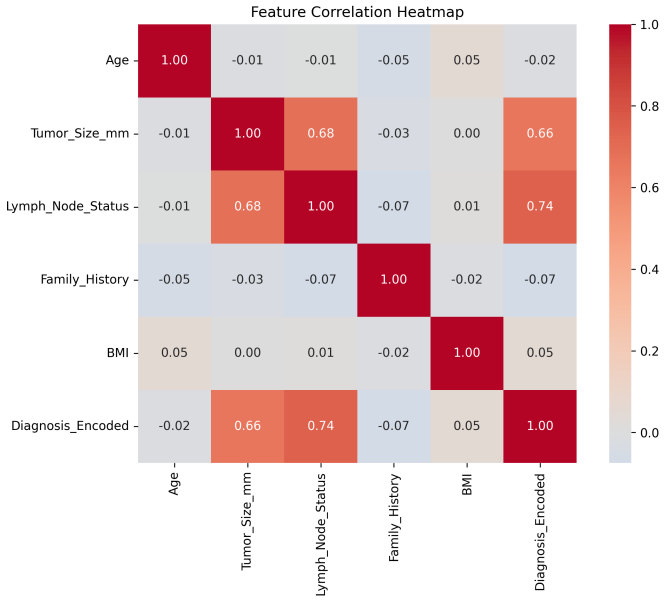


Fig. 4. Correlation heatmap of the selected dataset features.

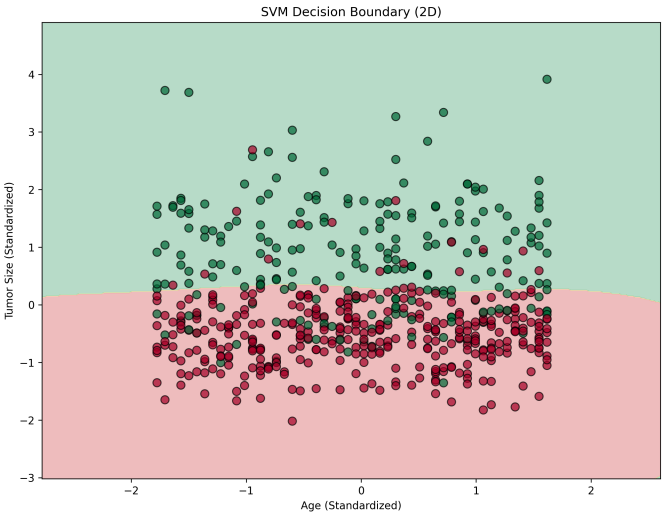


Fig. 5. Two-dimensional SVM decision boundary using Age and Tumor Size.

TABLE II
PERFORMANCE COMPARISON OF CLASSIFICATION MODELS

Metric	SVM (RBF)	Decision Tree
Accuracy	**88.60%**	84.21%
Precision	**81.63%**	78.26%
Recall	**90.91%**	81.82%
F1-score	**86.02%**	80.00%

The SVM classifier achieved an accuracy of 88.60%, which is higher than the 84.21% accuracy obtained by the Decision Tree baseline.

The SVM model also achieved higher precision, recall, and F1-score on the test dataset. Its recall of 90.91% indicates that it correctly identified 40 of the 44 malignant records in the test set.

The Decision Tree classifier achieved an accuracy of 84.21%. Although it provides a useful baseline for comparison, its overall performance was lower than that of the SVM classifier in this experiment.

These results show that the RBF-based SVM performed better than the selected Decision Tree baseline on this particular synthetic dataset and train-test split.

VIII. LIMITATIONS

The dataset used in this study is synthetic and does not contain real patient records collected from hospitals or clinical institutions. Therefore, the experimental results cannot be interpreted as evidence of clinical accuracy or real-world diagnostic effectiveness.

The dataset contains a limited number of records and only five input features are used for classification. Further evaluation using independently collected real-world data, larger datasets, and clinical validation would be required before considering any practical healthcare application.

IX. CONCLUSION

This study presented a machine learning-based approach for classifying breast tumor records using Support Vector Machine with an RBF kernel. A custom synthetic dataset containing 569 records representing six regions of Jharkhand was used for the experiment.

The SVM classifier achieved 88.60% accuracy, while the Decision Tree baseline achieved 84.21% accuracy. The SVM model also obtained higher precision, recall, and F1-score than the baseline model on the selected test split.

The results demonstrate the performance of the SVM classifier on the synthetic dataset used in this experiment. However, the findings should not be considered clinical validation. Testing with real patient data and independent clinical evaluation remain necessary for future research.

X. FUTURE SCOPE

Future work may focus on the following improvements:

- Evaluating the proposed model using real and anonymized patient data collected with appropriate institutional and ethical approvals.
- Increasing the dataset size and including additional clinically relevant features.
- Applying cross-validation and systematic hyperparameter optimization.
- Developing a web-based interface for experimental demonstration.
- Conducting independent validation before considering any clinical application.

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